Chapter 9: One- and Two-Sample Estimation Review

Agenda

- Review of critical values and quantiles.
- Computing z, t, and χ^2 confidence intervals for normal data.
- Conceptual view of confidence intervals.
- Confidence intervals for polling (Bernoulli distributions).
- CLT $\Rightarrow$ large-sample confidence intervals for the mean.

Review of Critical Values and Quantiles

- Quantile: For a distribution, $q(f)$ is a value such that a fraction f of the distribution lies at or below $q(f)$. left-tail probability

$$P(X \leq q(f)) = f$$

- Common notation:
  - z_α for $N(0,1)$
  - t_α for Student's t
  - χ^2_α for the chi-square distribution
  - F_α for the F distribution

Concept Question

1. $z_{0.025} =$
   (a) -1.96
   (b) -0.95
   (c) 0.95
   (d) 1.96
   (e) 2.87

.975

2. $-z_{0.16} =$
   (a) -1.33
   (b) -0.99
   (c) 0.99
   (d) 1.33
   (e) 3.52

Computing Confidence Intervals from Normal Data

Suppose $x_1, \ldots, x_n$ are drawn from $N(\mu, \sigma^2)$, and the confidence level is $1 - \alpha$.

z-confidence interval for μ (σ known):

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

t-confidence interval for μ (σ unknown):

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

χ^2-confidence interval for σ^2:

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2}}, \ \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}} \right)$$

Both t and χ^2 use $n - 1$ degrees of freedom.

z Rule of Thumb

Suppose $x_1, \ldots, x_n \sim N(\mu, \sigma^2)$ with σ known.

Rule-of-thumb 95% confidence interval for μ:

$$\bar{x} \pm 2\frac{\sigma}{\sqrt{n}}$$

More precise 95% confidence interval for μ:

$$\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

Computing Confidence Intervals

$\bar{x} = 2.5$

$z_{.05} = 1.645$

The data $1, 2, 3, 4$ is drawn from $N(\mu, \sigma^2)$ with μ unknown.

$2.5 \pm 1.645 = (0.855, 4.1)$

1. Find a 90% z-confidence interval for μ, given that $\sigma = 2$.
2. For the remaining parts, suppose σ is unknown. Find a 90% t-confidence interval for μ.

   $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{5}{3}$; $s = \sqrt{\frac{5}{3}}$; $t_{0.05,3} = 2.35$

   $CI = 2.5 \pm 1.52 = (.98, 4.0)$
3. Find a 90% χ^2-confidence interval for σ^2.
4. Find a 90% χ^2-confidence interval for σ.
5. Given a normal sample with $n = 100$, $\bar{x} = 12$, and $s = 5$, find the rule-of-thumb 95% confidence interval for μ.

3. $\left(\frac{(n-1)s^2}{\chi^2_{.05}}, \frac{(n-1)s^2}{\chi^2_{.95}} \right) = \left(\frac{5}{7.815}, \frac{5}{.352} \right) = (.64, 14.2)$

4. $(.8, 3.7)$

5. $CI = 12 \pm 2\left(\frac{5}{\sqrt{100}}\right) = 12 \pm 1$

Conceptual View of Confidence Intervals

- Computed from data $\Rightarrow$ interval statistic.
- Used to estimate a parameter $\Rightarrow$ interval estimate.
- Width and confidence level measure precision and performance.
- Confidence intervals are a **frequentist** method.

Important interpretation:

- A 95% confidence interval of $[1.2, 3.4]$ for μ does *not* mean

$$P(1.2 \leq \mu \leq 3.4) = 0.95.$$

- Instead, the random interval covers the true parameter in 95% of repeated samples.

Discussion: What Affects the Width?

How does the width of a confidence interval for the mean change if:

1. we increase n?
2. we increase the confidence level?
3. we increase μ?
4. we increase σ?

B

A

Choose:

(A) it gets wider

(B) it gets narrower

(C) it stays the same

Polling: Binomial Proportion Confidence Interval

Suppose $X_1, \ldots, X_n$ are Bernoulli(p) with p unknown.
A normal $(1-\alpha)$ confidence interval for p is

$$\bar{x} \pm z_{\alpha/2} \frac{1}{2\sqrt{n}}.$$

This uses:

- the CLT
- the bound

$$\sqrt{p(1-p)} \leq \frac{1}{2}.$$

For 95% confidence, a common polling margin of error is approximately

$$\bar{x} \pm \frac{1}{\sqrt{n}}.$$

So a margin of error of ± 0.05 corresponds to about

$$n = 400.$$

Polling

A $(1-\alpha)$ confidence interval for p is

$$\bar{x} \pm z_{\alpha/2}\frac{1}{2\sqrt{n}}.$$

1. How many people would you have to poll to have a margin of error of 0.01 with 95% confidence?
2. How many people would you have to poll to have a margin of error of 0.01 with 80% confidence?

Non-Normal Data: Large-Sample Confidence Intervals

Suppose $x_1, \ldots, x_n$ are drawn from a distribution that may not be normal, but has finite mean and variance.
A version of the CLT says that for large n,

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} \approx N(0, 1).$$

Therefore, for large n, an approximate $(1 - \alpha)$ confidence interval for μ is

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

This is called the **large-sample confidence interval**.